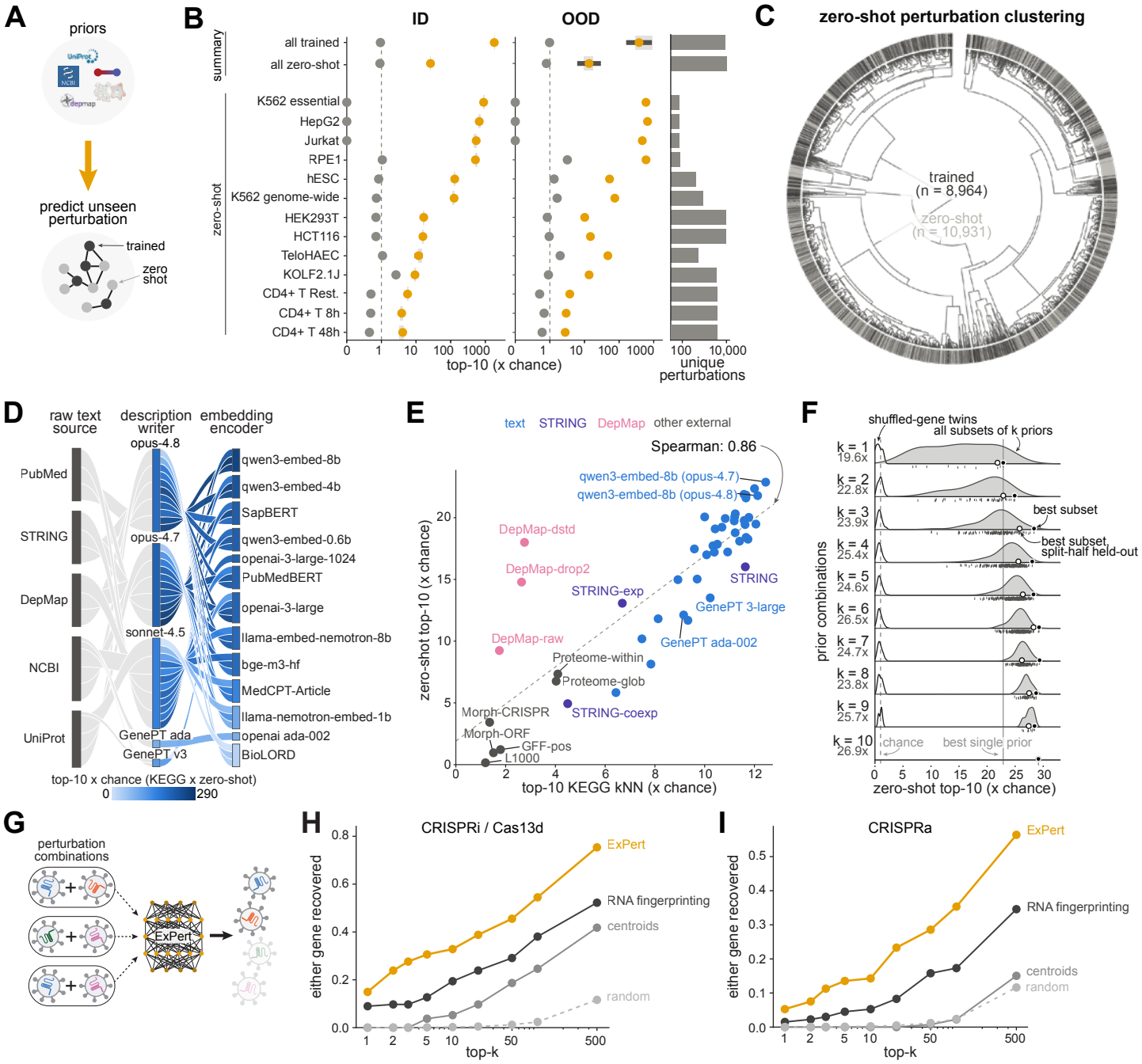
Figure 3


A
priors
predict unseen perturbation
trained
zero shot
B
ID
OOD
summary
all trained
all zero-shot
zero-shot
K562 essential
HepG2
Jurkat
RPE1
hESC
K562 genome-wide
HEK293T
HCT116
TeloHAEC
KOLF2.1J
CD4+ T Rest.
CD4+ T 8h
CD4+ T 48h
top-10 (x chance)
unique perturbations
C
zero-shot perturbation clustering
trained (n = 8,964)
zero-shot (n = 10,931)
D
raw text source
description writer
embedding encoder
PubMed
STRING
DepMap
NCBI
UniProt
opus-4.8
opus-4.7
sonnet-4.5
GenePT ada
GenePT v3
qwen3-embed-8b
qwen3-embed-4b
SapBERT
qwen3-embed-0.6b
openai-3-large-1024
PubMedBERT
openai-3-large
llama-embed-nemotron-8b
bge-m3-hf
MedCPT-Article
llama-nemotron-embed-1b
openai ada-002
BioLORD
top-10 x chance (KEGG x zero-shot)
0
290
E
text
STRING
DepMap
other external
Spearman: 0.86
qwen3-embed-8b (opus-4.7)
qwen3-embed-8b (opus-4.8)
DepMap-dstd
DepMap-drop2
STRING
STRING-exp
GenePT 3-large
GenePT ada-002
DepMap-raw
Proteome-within
Proteome-glob
Morph-CRISPR
STRING-coexp
Morph-ORF
GFF-pos
L1000
zero-shot top-10 (x chance)
top-10 KEGG kNN (x chance)
F
shuffled-gene twins
all subsets of k priors
k = 1 19.6x
k = 2 22.8x
k = 3 23.9x
k = 4 25.4x
k = 5 24.6x
k = 6 26.5x
k = 7 24.7x
k = 8 23.8x
k = 9 25.7x
k = 10 26.9x
best subset
best subset, split-half held-out
chance
best single prior
prior combinations
zero-shot top-10 (x chance)
G
perturbation combinations
ExPert
H
CRISPRi / Cas13d
ExPert
RNA fingerprinting
centroids
random
either gene recovered
top-k
I
CRISPRa
ExPert
RNA fingerprinting
centroids
random
either gene recovered
top-k